

DAY 2 • OPENING DEMO • 10 MINUTES

Kustomize

One base, many ships

Yesterday you each pushed a raft. Overnight I changed all of them with one file. Watch.



What just happened to your raft

- ♦ Yesterday at end-of-day, every student's namespace had `raft-fleet/<your-name>:v1` running as a Pod.
- ♦ This morning, every namespace is running `<your-name>:v2` — different background, different message.
- ♦ I did not touch your namespace. I touched **one file**.
- ♦ That file is called a **Kustomize overlay**, and it is the single most useful operator's tool in Kubernetes.

Pull up `k9s` and confirm. Your Pod is on v2.

Two different problems

Problem	Tool
Same shape, different <i>parameters</i> (image tag, namespace, replicas) for many copies	Kustomize — patches over a base
Same shape, packaged as a <i>product</i> you hand someone else to install	Helm — templates + values (tomorrow)

- ♦ Kustomize is the tool the **operator** reaches for.
- ♦ Helm is the tool the **publisher** reaches for.
- ♦ Today you only need to recognise the shape of Kustomize — so when ArgoCD reads its overlays on Day 3, it isn't a surprise. Helm you'll *write* tomorrow.

The shape of what I ran

```
kustomize-demo/  
├── base/  
│   └── raft-pod.yaml      ← the Day 1 Pod (image is a placeholder)  
└── overlays/  
    └── all-students/  
        └── kustomization.yaml ← one images: override, rewritten per student
```

```
# overlays/all-students/kustomization.yaml (excerpt)  
images:  
  name: harbor.wagbiz.org/raft-fleet/raft # placeholder in base  
  newName: harbor.wagbiz.org/raft-fleet/raft # rewritten per student at apply  
  newTag: v2
```

One overlay shape. The loop parameterises its image per student.

The loop I ran at 7am

```
for ns in $(kubectl get ns -l role=student -o name); do
  student= ${ns#namespace/}
  ( cd overlays/all-students && \
    kustomize edit set image \
      "harbor.kodai.com:5000/raft/raft-harbor.wegala.org/raft-fluent:${student}-v2" && \
    kubectl apply -k . -n ${ns#namespace/} )
done
```

- ♦ For each student namespace, `kustomize edit set image` rewrites the overlay's image override to that student's v2 repo.
- ♦ `kubectl apply -k .` updates the existing `my-raft` Pod in-place — same Pod, new image.
- ♦ **Wall-clock: ~4 seconds.** Cluster-wide.

This is how I will give you each a Deployment in 10 minutes from now. And a Service this afternoon. And a 3-tier app this afternoon.

Now: meet Captain Kube.

That's all you need to know about Kustomize today. The lecture starts now.

